

CC 352

Computer Graphics

Lecture 8

Design of Spline Algorithms

Spline representations

A **spline** is a smooth curve defined mathematically using a piecewise polynomial curves used in computer graphics to represent smooth and intricate shapes

Spline representations are a powerful tool for creating smooth and flexible curves. They are formed by piecing together polynomial segments of varying degrees, connected at points called **control points**. These control points define the overall shape of the curve or surface, and manipulating them allows you to achieve a wide range of effects.

key points

- ▶ Splines can be either **interpolating** or **approximating**.
- ❑ Interpolating splines *pass exactly* through the specified control points,
- ❑ Approximating splines *come close to* the control points but don't necessarily pass through them.
- ▶ The optimal degree for a spline depends on the specific application and desired balance of smoothness, complexity, and computational efficiency

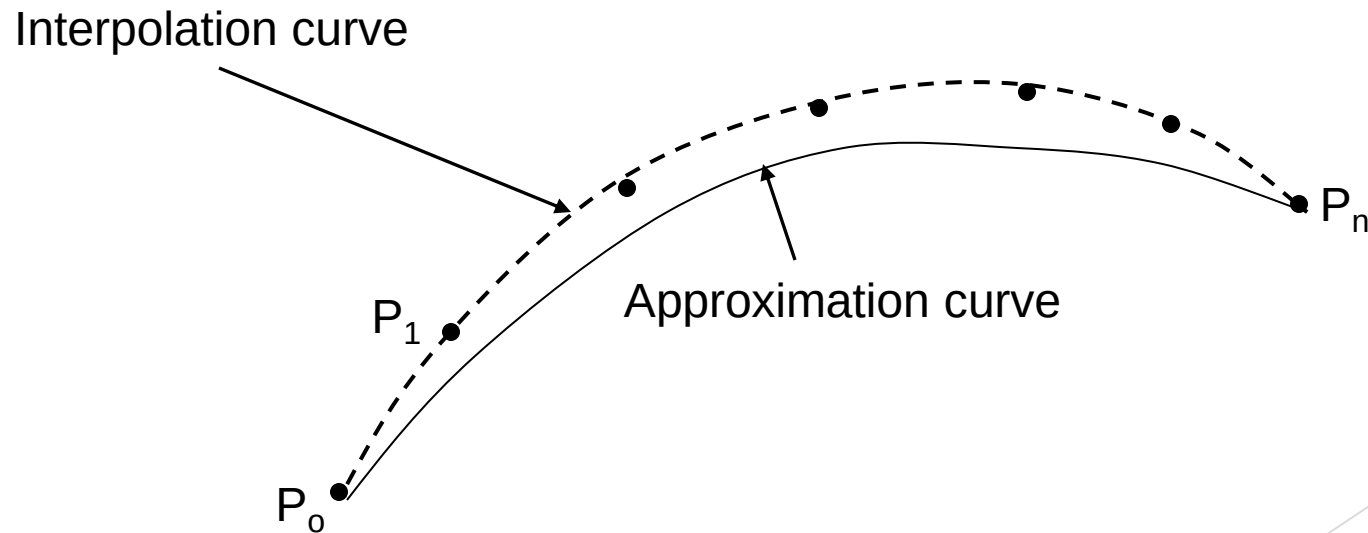


APPROXIMATION SPLINE

- Does not necessarily pass through all control points
- General representation is

$$P(u) = \sum F_i(n,u) P_i$$

where- F_i is the blending function.
 n is no. of control points.



Types of Curves / Surfaces

- Explicit:

$$y = mx + b$$

$$z = A x + B y + C$$

- Implicit:

$$Ax + By + C = 0$$

$$(x - x_0)^2 + (y - y_0)^2 - r^2 = 0$$

- Parametric:

$$x = x_0 + (x_1 - x_0)t$$

$$x = x_0 + r\cos\theta$$

$$y = y_0 + (y_1 - y_0)t$$

$$y = y_0 + r\sin\theta$$

Types of Curves / Surfaces

- Implicit Surfaces
 - Metaball
 - **Parametric surfaces**
 - Hermite curve
 - Bezier curve
 - Biubic patches
 - Tessalation
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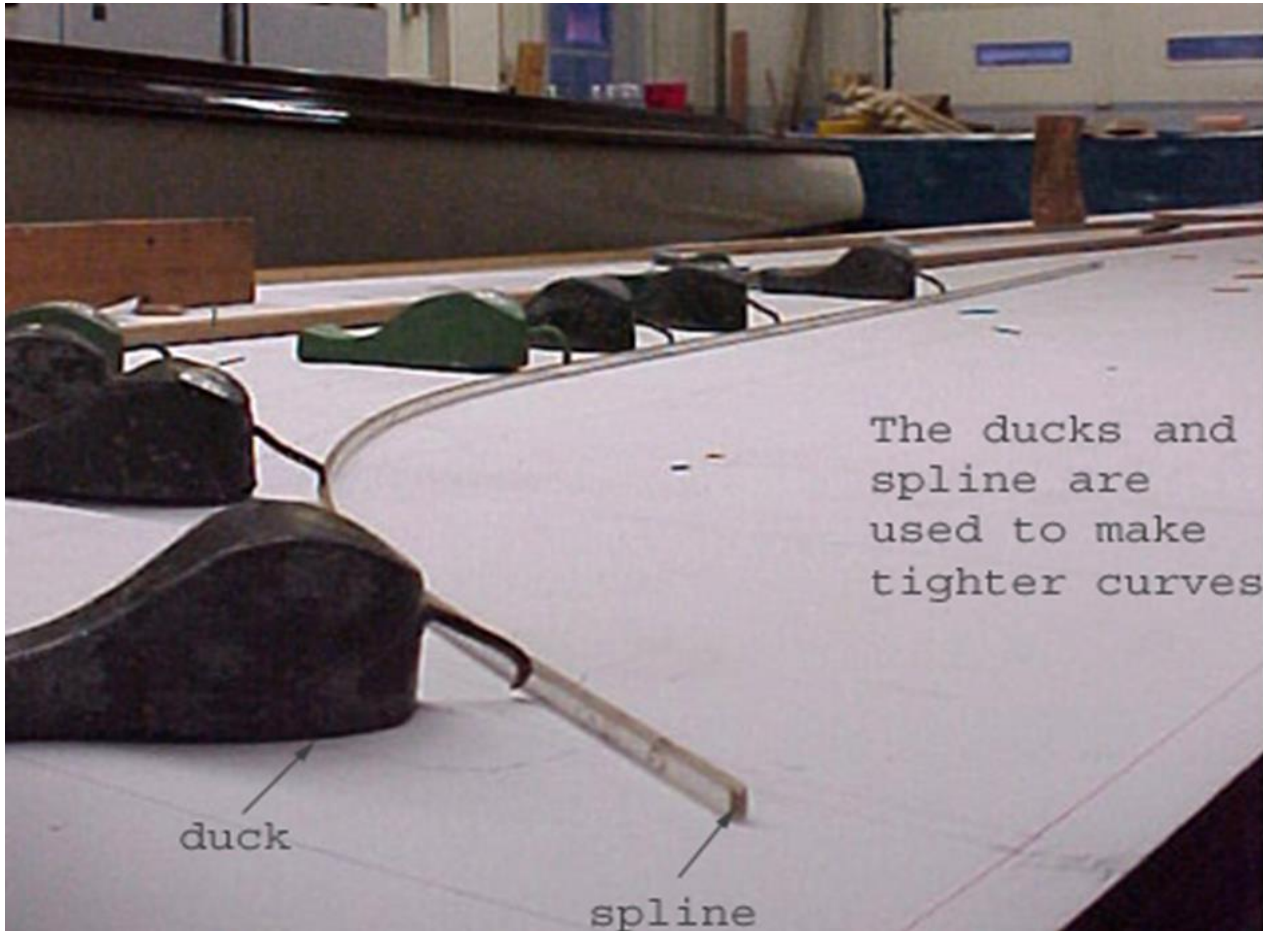
Why parametric?

- Parametric curves are very flexible.
- Parameter count gives the object's dimension.

$(x(u,v), y(u,v), z(u,v))$: 2D surface

- Coord functions independent.
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Bézier Spline Curves



Pierre Bézier

General formula for Bezier curve

- ▶ $B(t) = \sum_{i=0}^n P_i b_{i,n}(t)$ where $0 \leq t \leq 1$
- ▶ $B(t)$: Bezier curve
- ▶ n : the degree of the curve
- ▶ P_i : The control points
- ▶ $B_{i,n}$: The base function which use Bernstein polynomials with the general form:

$$\beta_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}$$

$$b_{0,0}(t) = 1$$

$$b_{0,1}(t) = 1 - t$$

$$b_{0,2}(t) = (1 - t)^2$$

$$b_{0,3}(t) = (1 - t)^3$$

$$b_{1,1}(t) = t$$

$$b_{1,2}(t) = 2t(1 - t)$$

$$b_{2,2}(t) = t^2$$

$$b_{1,3}(t) = 3t(1 - t)^2$$

$$b_{2,3}(t) = 3t^2(1 - t)$$

$$b_{3,3}(t) = t^3$$

n = 2

We have P_0, P_1, P_2

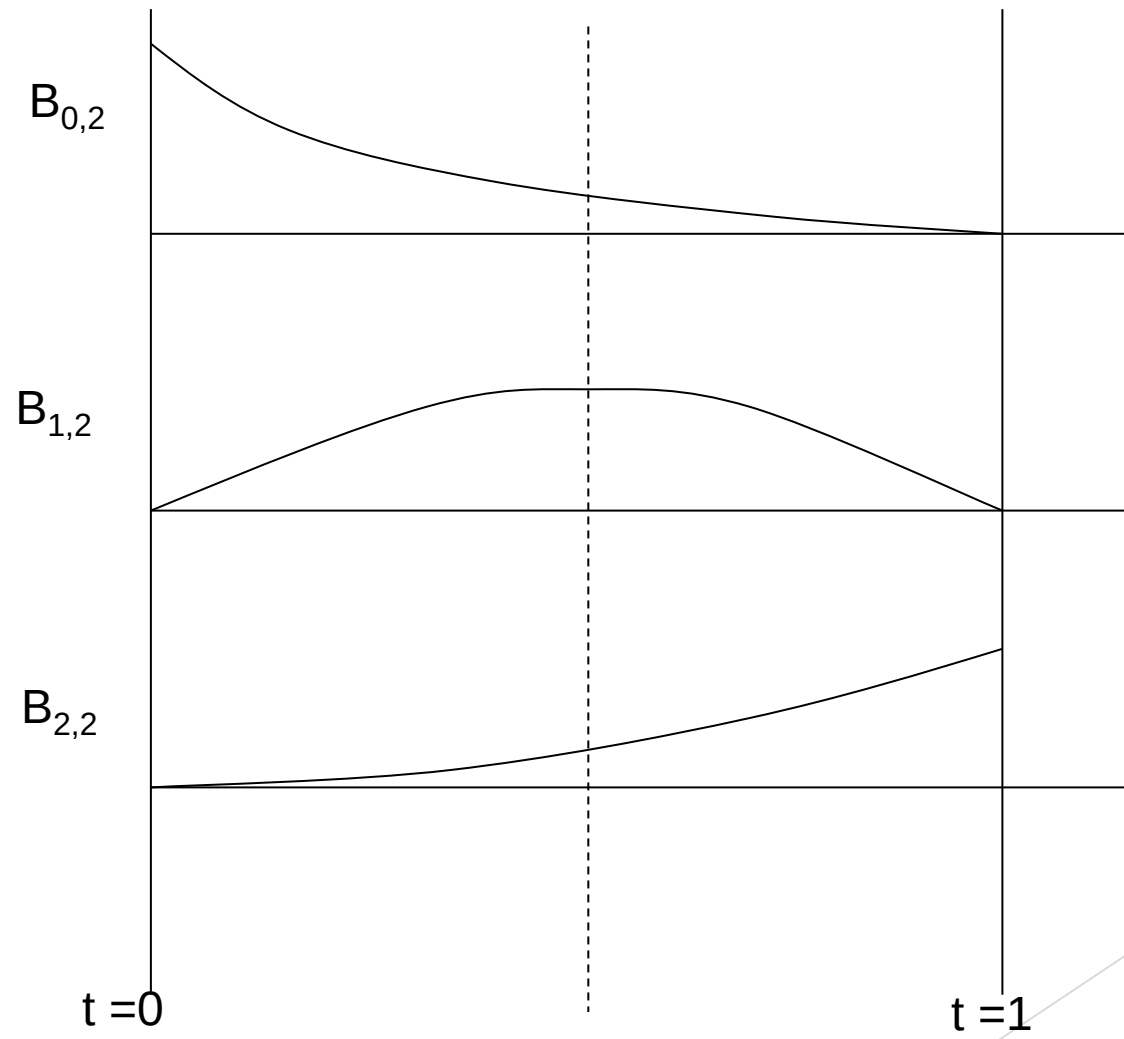
- $B_{0,2} = {}^2C_0 t^0 (1-t)^2$
 $= (1-t)^2$

- $B_{1,2} = {}^2C_1 t^1 (1-t)^1$
 $= 2t(1-t)$

- $B_{2,2} = {}^2C_2 t^2 (1-t)^0$
 $= t^2$

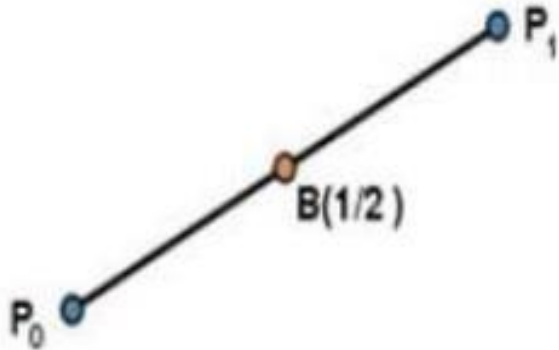
$$p(t) = (1-t)^2 P_0 + 2t(1-t)P_1 + t^2 P_2$$

VARIATION OF BLENDING FUNCTIONS

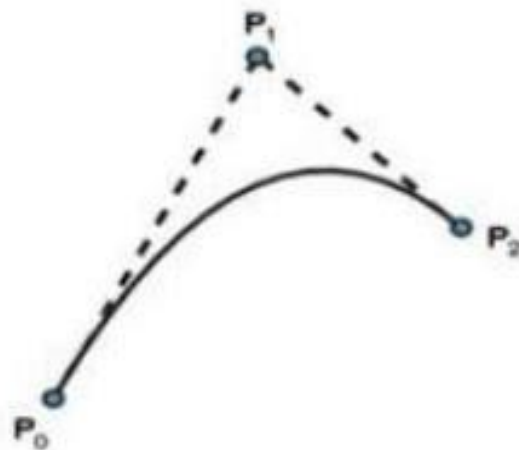


Different degrees of Bézier curves:

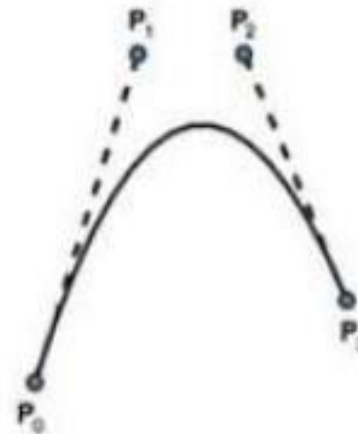
The simplest Bézier curve is the straight line from the point P_0 to P_1 . A quadratic Bézier curve is determined by three control points. A cubic Bézier curve is determined by four control points.



Simple Bézier Curve



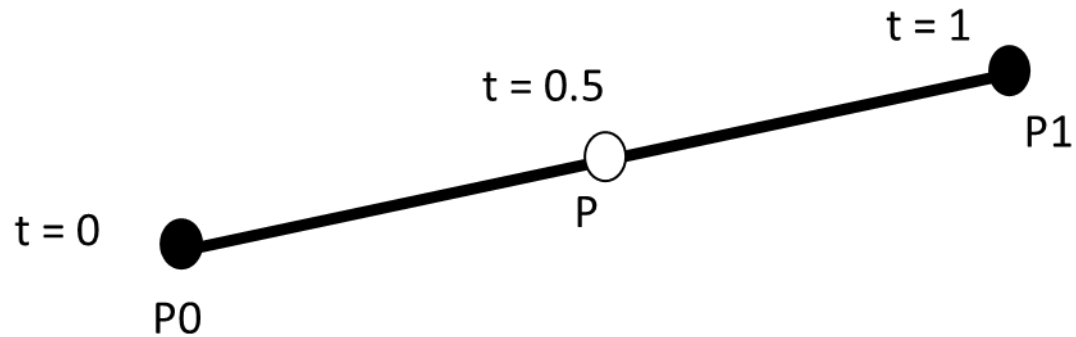
Quadratic Bézier Curve



Cubic Bézier Curve

Degree 1 (linear) Bezier

- ▶ A parameter t is used to determine the value of the variables
- ▶ $X(t) = (1-t)*X_0 + t*X_1$
- ▶ $Y(t) = (1-t)*Y_0 + t*Y_1$ Where $0 < t < 1$.
- ▶ Let $P_0 = (X_0, Y_0)$ and $P_1 = (X_1, Y_1)$ and $P = (X, Y)$ then
- ▶ **$P(t) = (1-t)*P_0 + t*P_1$**



Degree 2 (quadratic) Bezier

- ▶ A parameter t is used to determine the value of the variables
- ▶ $X(t) = (1-t)^2X_0 + 2(1-t)tX_1 + t^2X_2$
- ▶ $Y(t) = (1-t)^2Y_0 + 2(1-t)tY_1 + t^2Y_2$ Where $0 < t < 1$.
- ▶ Let $P_0 = (X_0, Y_0)$, $P_1 = (X_1, Y_1)$, $P_2 = (X_2, Y_2)$ and $P = (X, Y)$ then
- ▶ **$P(t) = (1-t)^2P_0 + 2(1-t)tP_1 + t^2P_2$**

Degree 3 (Cubic) Bezier

- ▶ A parameter t is used to determine the value of the variables
- ▶ $X(t) = (1-t)^3X_0 + 3(1-t)^2tX_1 + 3(1-t)t^2X_2 + t^3X_3$
- ▶ $Y(t) = (1-t)^3Y_0 + 3(1-t)^2tY_1 + 3(1-t)t^2Y_2 + t^3Y_3$ Where $0 < t < 1$.
- ▶ Let $P_0 = (X_0, Y_0)$, $P_1 = (X_1, Y_1)$, $P_2 = (X_2, Y_2)$, $P_3 = (X_3, Y_3)$ and $P = (X, Y)$ then
- ▶ **$P(t) = (1-t)^3P_0 + 3(1-t)^2tP_1 + 3(1-t)t^2P_2 + t^3P_3$**

Example 1:

- Given the three control points:

$$P_0 = (2, 2)$$

$$P_1 = (4, 6)$$

$$P_2 = (9, 3)$$

Plot the Bezier curve along with the three control points.

► The Quadratic Bezier curve is given by:

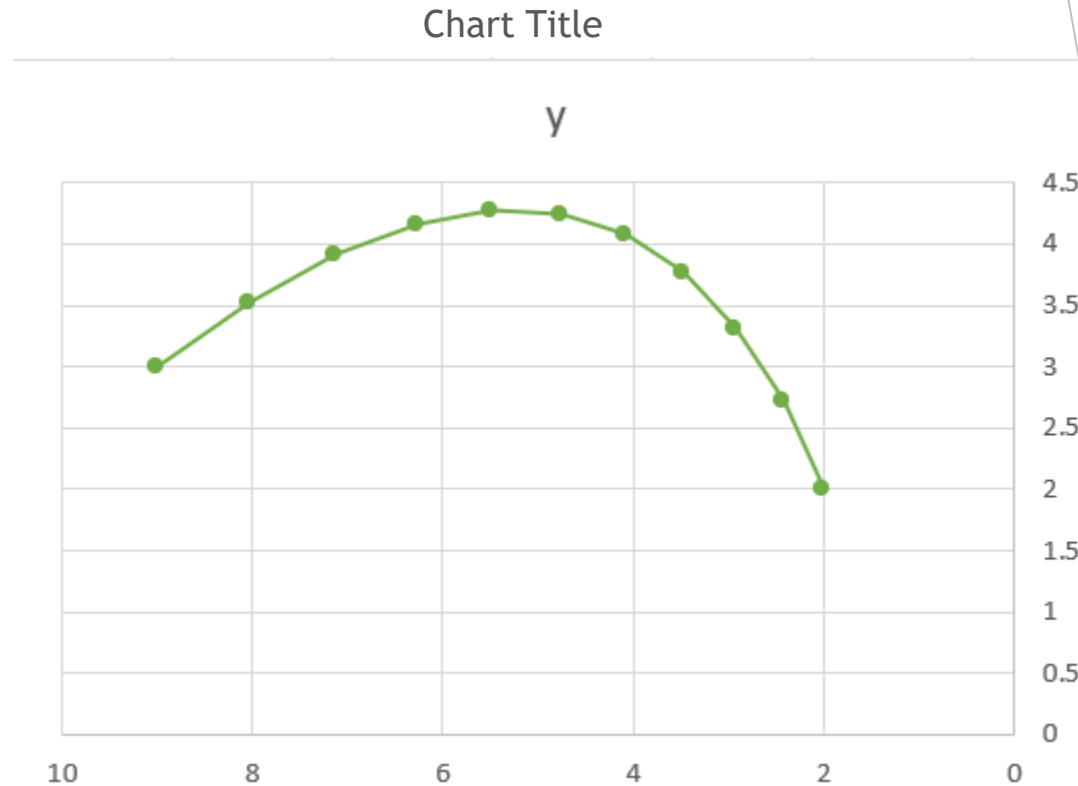
► $X(t) = (1-t)^2X_0 + 2(1-t)tX_1 + t^2X_2$

► $Y(t) = (1-t)^2Y_0 + 2(1-t)tY_1 + t^2Y_2$ Where $0 < t < 1$.

► Let $P_0 = (X_0, Y_0)$, $P_1 = (X_1, Y_1)$, $P_2 = (X_2, Y_2)$ and $P = (X, Y)$ then

► $P(t) = (1-t)^2P_0 + 2(1-t)tP_1 + t^2P_2$

T	X	Y
0	2	2
0.1	2.43	2.73
0.2	2.92	3.32
0.3	3.47	3.77
0.4	4.08	4.08
0.5	4.75	4.25
0.6	5.48	4.28
0.7	6.27	4.17
0.8	7.12	3.92
0.9	8.03	3.53
1	9	3



Example 2:

- ▶ Given the four control points:

$$P_0 = (1, 1)$$

$$P_1 = (2, 8)$$

$$P_2 = (6, 0)$$

$$P_3 = (8, 7)$$

- ▶ Plot the Bezier curve along with the four control points.

► The cubic Bezier curve is given by:

$$P(t) = (1-t)^3 \cdot P_0 + 3 \cdot (1-t)^2 \cdot t \cdot P_1 + 3 \cdot (1-t) \cdot t^2 \cdot P_2 + t^3 \cdot P_3$$

$$= (1-t)^3 (1, 1) + 3t(1-t)^2 (2, 8) + 3t^2(1-t)(6, 0) + t^3 (8, 7)$$

$$= (-5t^3 + 9t^2 + 3t + 1, 30t^3 - 45t^2 + 21t + 1)$$

T	X	Y
0	1	1
0.1	1.385	2.68
0.2	1.92	3.64
0.3	2.575	4.06
0.4	3.32	4.12
0.5	4.125	4
0.6	4.96	3.88
0.7	5.795	3.94
0.8	6.6	4.36
0.9	7.345	5.32
1	8	7



THANK YOU